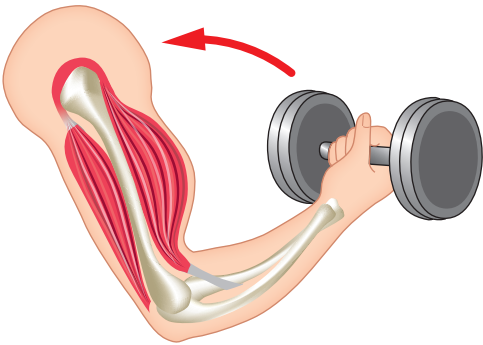


Option B: Human Musculoskeletal Anatomy

8. **Image 8.1** shows a weight being lifted by a human forelimb.

Image 8.1



- (a) (i) With reference to the names of the muscles involved, state how the muscles work together to lift the weight. [1]

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- (ii) **Image 8.1** represents a third order lever. Use **Image 8.1** and your knowledge of levers to identify the effort, load and fulcrum and explain why it is a third order lever. [2]

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Another experiment was carried out into the rate of muscle fatigue when the weight was held in the position as shown in **Image 8.1**.

- (b) (i) Suggest what causes the muscle to fatigue. [2]

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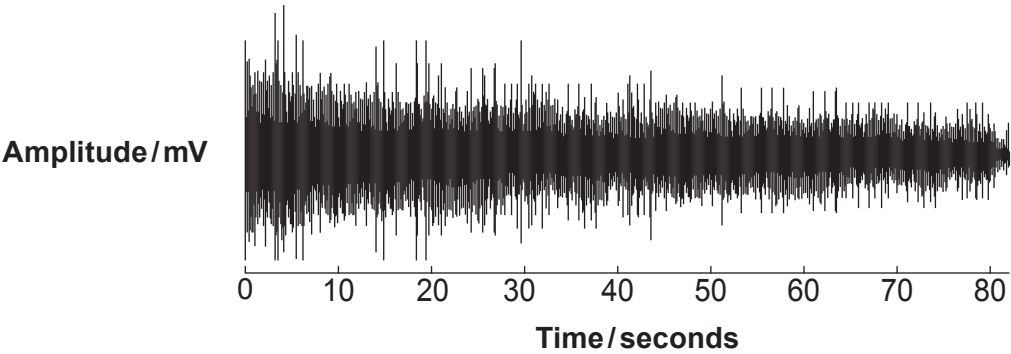
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Image 8.2 shows an electromyogram (EMG) which is used to measure fatigue over time. The amplitude (mV) of the signal indicates the force the muscle is generating.

Image 8.2



An EMG was produced for three males and three females. The total time they could hold the weight and the mean amplitude for the first five seconds and last five seconds of the trace was recorded for each person. The results are shown in **Table 8.3**.

Table 8.3

Gender	Name	Total time /s	Mean amplitude for first five seconds /mV	Mean amplitude for last five seconds /mV	Rate of fatigue
male	Robert	85	26.85	17.20	−0.11
	George	135	29.13	5.03	−0.18
	Gabriel	90	6.83	2.10	
female	Mary	230	9.32	3.48	−0.03
	Elizabeth	120	14.31	4.09	−0.09
	Alexandra	135	12.11	3.55	−0.06

The rate of fatigue is measured using the equation:

Rate of fatigue =
$$\frac{\text{mean amplitude for last five seconds} - \text{mean amplitude for first five seconds}}{\text{time}}$$

- (ii) Use the equation to calculate the rate of fatigue for **Gabriel**. Suggest a unit for rate of fatigue. [3]

Rate of fatigue =

Unit =



- (iii) It was concluded that females had better endurance than males. With reference to the data in **Table 8.3** comment on the validity of this conclusion. [3]

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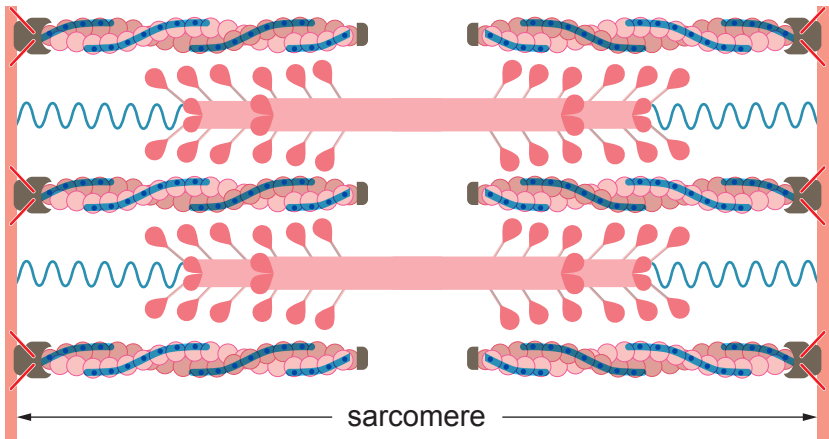
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- (c) **Image 8.4** shows a sarcomere.

Image 8.4



- (i) Use the letters **X**, **Y** and **Z** to label the following proteins on the sarcomere in **Image 8.4**. [1]

- X** actin
- Y** myosin
- Z** tropomyosin



- (ii) Low blood calcium increased the rate of muscle fatigue. Use your knowledge of sliding filament theory to explain the reason for this. [4]

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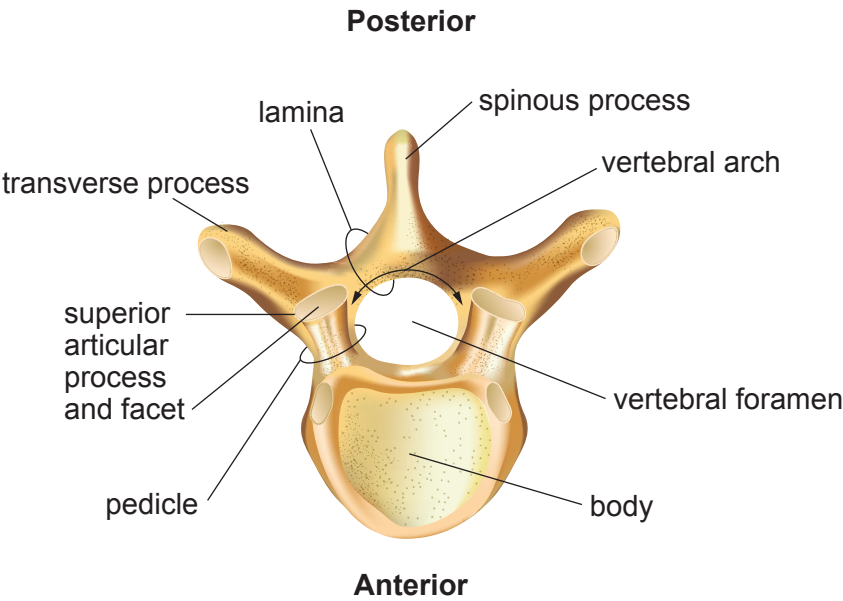
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- (d) **Image 8.5** shows a vertebra.

Image 8.5



- (i) State what spinal region the vertebra in **Image 8.5** is from and explain your answer. [2]

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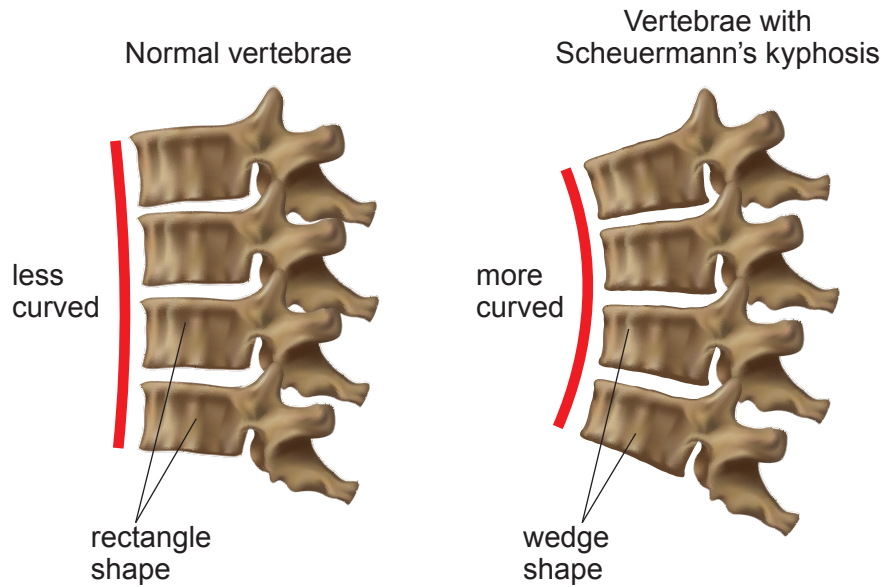
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Scheuermann's kyphosis is a medical condition similar to scoliosis which causes curvature of the spine due to vertebrae being wedge-shaped as shown in **Image 8.6**.

Image 8.6



(ii) Suggest which way the spine would curve in the body of a person with Scheuermann's kyphosis.

[1]

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(iii) Suggest a possible treatment for Scheuermann's kyphosis.

[1]

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